

EDUCATION

Courant Institute of Mathematical Sciences, New York University , computer science <i>PhD</i> . • Verifiable systems and theoretical computer science.	starting 2026.9
Stanford University , computer science <i>M.S.</i> and mathematics <i>B.A.</i> • General Yaowu Wang Scholarship.	2019.9 - 2024.6
St Paul’s School • Full scholarship.	2017.9 - 2019.6

PAPERS

- *Complex Cobordism and Formal Group Law*, Manuscript (Undergraduate Honours Thesis), Stanford University, 2024, Julius Zhang. pdf

EXPERIENCE

a16z • Design and build Jolt , an open-source zero-knowledge virtual-machine in Rust. • Proof compression research : designed torus compression system and proved theoretical optimality via Hilbert 90. • Audited elliptic-curve optimizations, formalizing folklore cryptographic results into rigorous proofs . • Performance engineering across the proving stack, including sum-check optimization.	Researcher, 2025.4 - Present
Millennium • Convex optimization and robust statistical estimation for real-time trading systems.	Researcher, 2024.6 - 2025.4
Stellar Development Foundation • Byzantine fault tolerance, C++, stellar-core . • Performance optimization: improved maximum transaction rate per second by 17% via changes to the consensus protocol, reducing number of communication rounds in overlay network. Formal verification in TLA+.	Consultant, 2023.6 - 2023.9
Distributed Systems and TCS Research, Stanford University • Under David Mazieres, research in general open membership Byzantine fault tolerance protocols (Github), with applications to certificate authority management and automatic password recovery with hardware security modules. • Research PCP under Aviad Rubinstein, presented full proof of the 2-to-2 Games Conjecture. Proved a result about Convex Continuous Class Complexity.	Researcher, 2023.9 - 2024.6
Topology Research, Stanford University • Honours thesis related to Floer homotopy theory under Mohammed Abouzaid (see Papers). Proved a result sketched by Daniel Quillen, Fields medalist, relating the Gysin homomorphism of complex cobordism to the Grothendieck residue symbol. Produced a recursive formula to compute quantum product correction term in Gromov-Witten invariant valued cohomology. • Research under Ciprian Manolescu. Floer theory, Seiberg-Witten, knot invariants. Produced new potential counterexamples for the smooth Poincare conjecture.	Researcher, 2020.7 - 2022.5

SKILLS

Computer Science: Rust, C/C++, Python, formal verification, convex optimization, distributed systems, cryptographic protocol design, performance engineering.

Mathematics: number theory, algebraic geometry, symplectic topology, Floer theory, gauge theory, concentration inequalities.